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School of Computer Science Engineering & Information Systems (SCORE)

LABFAT

Course code: ISWE406P

Course Title: Agile Development Process and Devops Lab

Lab Slot: L3+L4

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Creating Expense Tracking app Java Application in Local Machine

Folder Structure:

```
C:\projects\ExpenseTracker\
├──── src
│     ├──── main
│     │     └──── java
│     └──── test
│           └──── java
├──── target
│     ├──── classes
│     ├──── generated-sources
│     │     └──── annotations
│     ├──── generated-test-sources
│     │     └──── test-annotations
│     ├──── maven-archiver
│     ├──── maven-status
│     │     └──── maven-compiler-plugin
│     │           ├──── compile
│     │           │     └──── default-compile
│     │           └──── testCompile
│     │                 └──── default-testCompile
│     ├──── surefire-reports
│     └──── test-classes
├──── deployment.yaml
├──── Dockerfile
└──── pom.xml
```

ExpenseTracker.java

```
import java.util.*;

public class ExpenseTracker {

    private Map<String, String> users = new HashMap<>();
    private List<Expense> expenses = new ArrayList<>();

    public String registerUser(String username, String password) {
        if (users.containsKey(username)) {
            return "User already exists";
        }
        users.put(username, password);
        return "User registered successfully";
    }

    public String addExpense(String name, double price) {
        expenses.add(new Expense(name, price));
        return "Expense added";
    }

    public Expense findExpenseByName(String name) {
        for (Expense p : expenses) {
            if (p.getName().equalsIgnoreCase(name)) {
                return p;
            }
        }
        return null;
    }

    public String updateExpensePrice(String name, double newPrice) {
        for (Expense p : expenses) {
            if (p.getName().equalsIgnoreCase(name)) {
                p.setPrice(newPrice);
                return "Price updated";
            }
        }
        return "expense not found";
    }

    // Inner expense class
    static class Expense {
        private String name;
        private double price;

        public Expense(String name, double price) {
            this.name = name;
            this.price = price;
        }

        public String getName() { return name; }
        public double getPrice() { return price; }
```

```
        public void setPrice(double price) {
            this.price = price;
        }
    }
}
```

TodolistServiceTest.java

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class ExpenseTrackerTest {

    ExpenseTracker service = new ExpenseTracker();

    @Test
    void testRegisterUser() {
        String result = service.registerUser("john", "1234");
        assertEquals("User registered successfully", result);

        // Duplicate case
        String duplicate = service.registerUser("john", "1234");
        assertEquals("User already exists", duplicate);
    }

    @Test
    void testAddexpense() {
        String result = service.addExpense("Laptop", 50000);
        assertEquals("Expense added", result);
    }

    @Test
    void testFindExpense() {
        service.addExpense("Phone", 20000);

        ExpenseTracker.Expense expense = service.findExpenseByName("Phone");
        assertNotNull(expense);
        assertEquals(20000, expense.getPrice());
    }

    @Test
    void testUpdateExpensePrice() {
        service.addExpense("Tablet", 15000);

        String result = service.updateExpensePrice("Tablet", 18000);
        assertEquals("Price updated", result);

        ExpenseTracker.Expense updated = service.findExpenseByName("Tablet");
        assertEquals(18000, updated.getPrice());
    }
}
```

Maven file: pom.xml

```
    </dependencies>

    <build>
        <plugins>

            <!-- Compiler -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
            </plugin>

            <!-- Run tests -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.2.5</version>
                <configuration>
                    <useModulePath>false</useModulePath>
                </configuration>
            </plugin>

            <!-- Build runnable JAR -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.4.1</version>
                <configuration>
                    <archive>
                        <manifest>
                            <!-- 🔄 CHANGE: FULL path if inside package -->
                            <!-- e.g., com.app.AppService -->
                            <mainClass>ExpenseTracker</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>

        </plugins>
    </build>

</project>
```

Building Locally using Maven

```
PS C:\projects\ExpenseTracker> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.app:expense-tracker >-----------------------
[INFO] Building epense-tracker 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ expense-tracker ---
[INFO] Deleting C:\projects\ExpenseTracker\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ expense-tracker ---
[INFO] skip non existing resourceDirectory C:\projects\ExpenseTracker\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ expense-tracker ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 21] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ expense-tracker ---
[INFO] skip non existing resourceDirectory C:\projects\ExpenseTracker\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ expense-tracker ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ expense-tracker ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running ExpenseTrackerTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.072 s -- in ExpenseTrackerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ expense-tracker ---
[INFO] Building jar: C:\projects\ExpenseTracker\target\expense-tracker-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.095 s
[INFO] Finished at: 2026-04-13T10:31:34+05:30
[INFO] ------------------------------------------------------------------------
PS C:\projects\ExpenseTracker>
```

Creating Docker File

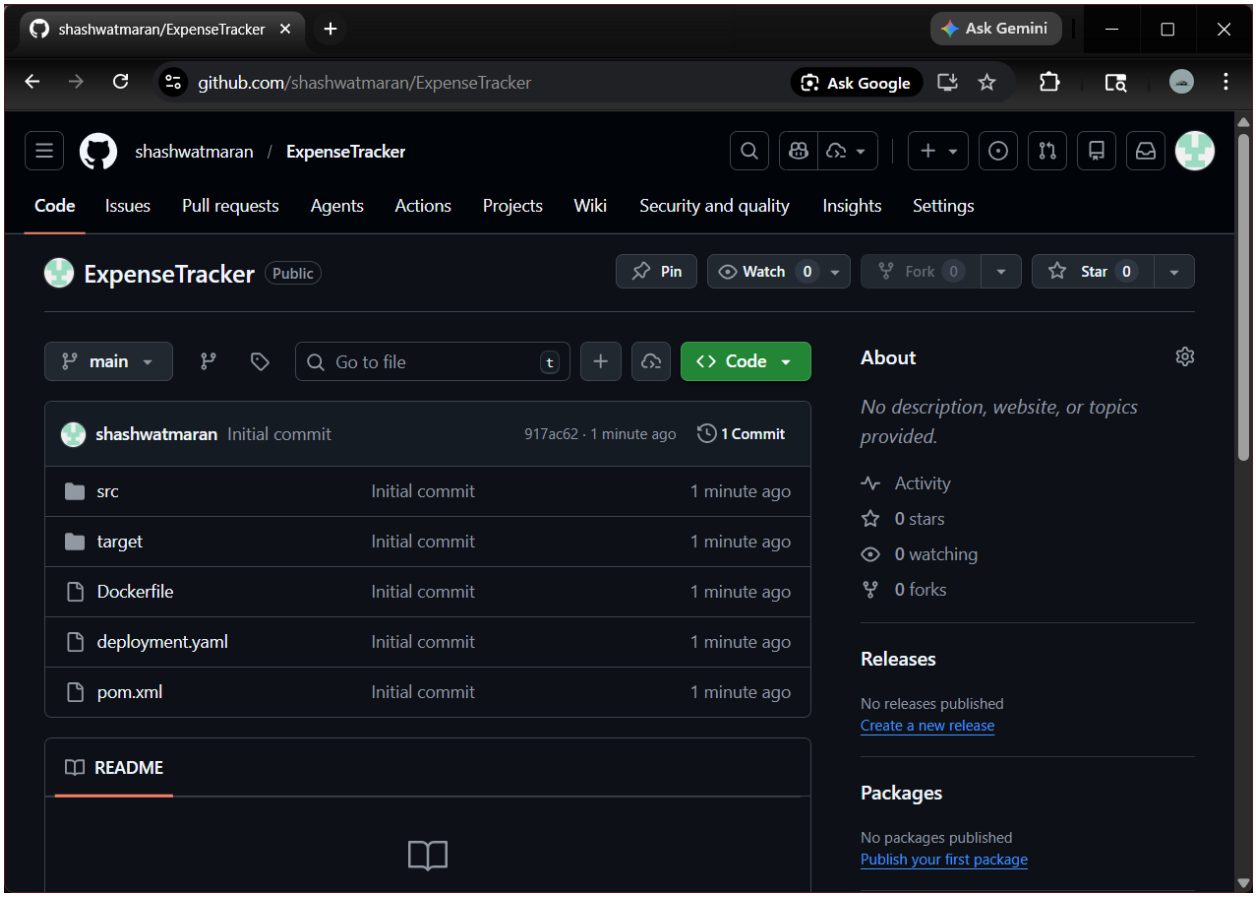
```
Dockerfile

FROM eclipse-temurin:21-jdk

COPY target/expense-tracker-1.0.jar app.jar

ENTRYPOINT ["java", "-jar", "app.jar"]
```

Kubernetes Deployment File

deployment.yaml

```yaml
! deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: expense-tracker

spec:
  replicas: 1
  selector:
    matchLabels:
      app: app
  template:
    metadata:
      labels:
        app: app

    spec:
      containers:
        - name: app-container

          image: shashmaran/expense-tracker:latest
          imagePullPolicy: Always

          ports:
            - containerPort: 8080

---
apiVersion: v1
kind: Service
metadata:
```

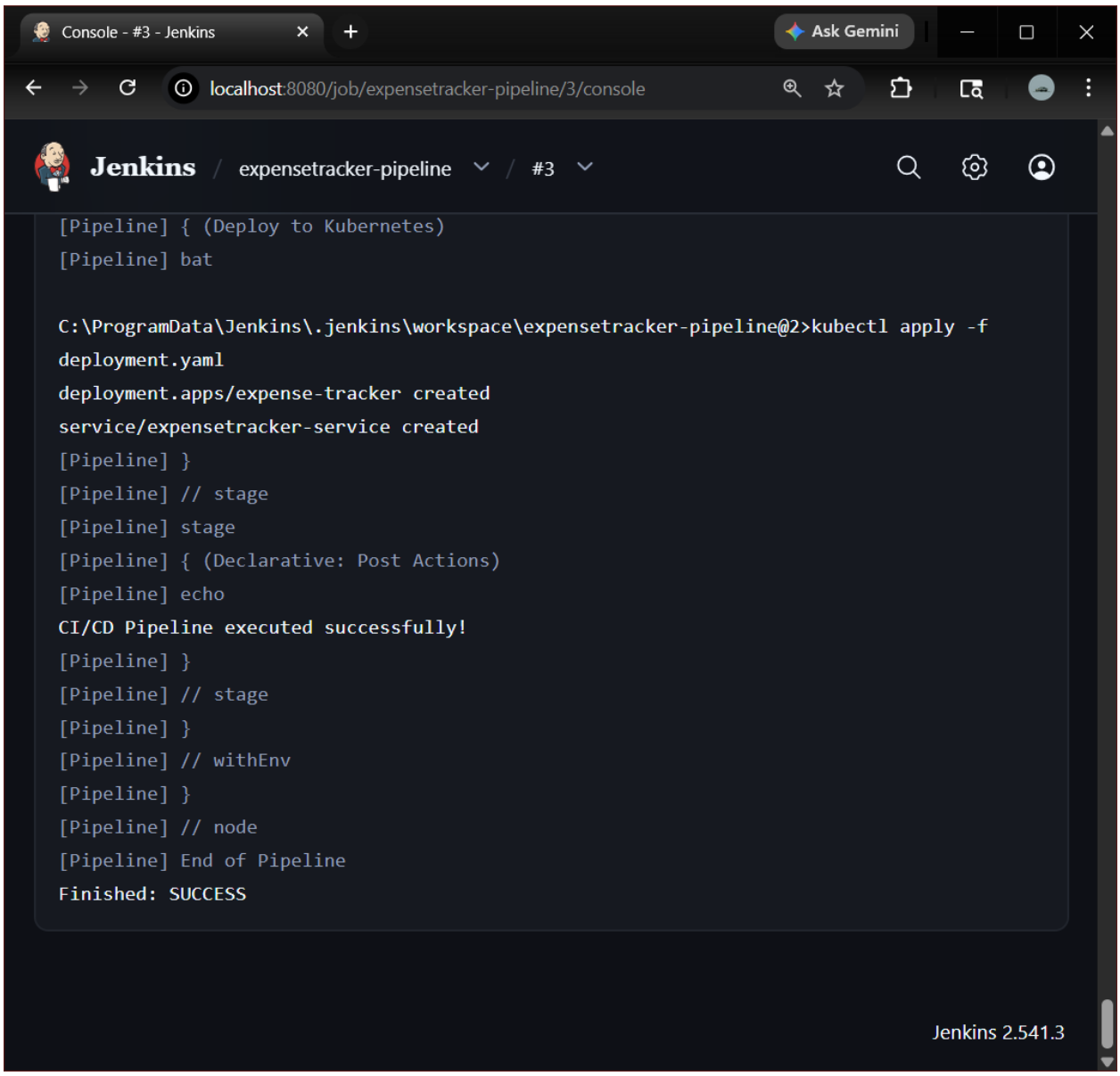
Pushing code to GitHub



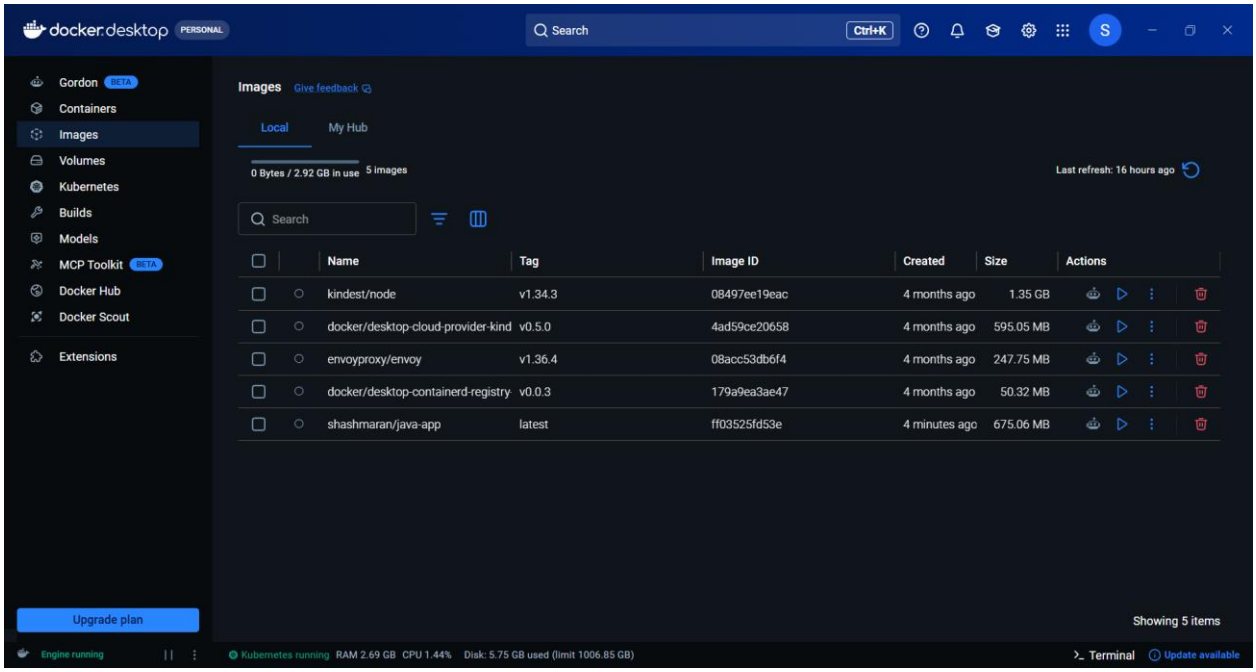
Jenkins Pipeline Job

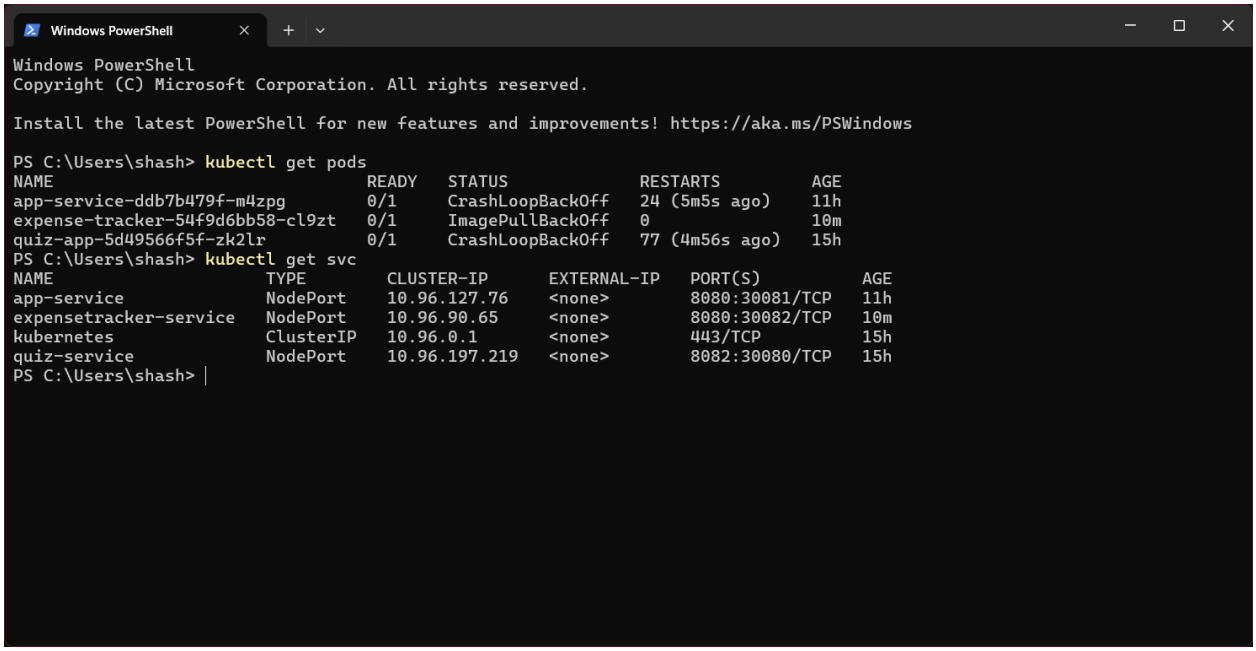
Pipeline script

Output:


Console - #3 - Jenkins
Ask Gemini
localhost:8080/job/expensetracker-pipeline/3/console
Jenkins
expensetracker-pipeline
#3
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\expensetracker-pipeline@2>kubectl apply -f deployment.yaml
deployment.apps/expense-tracker created
service/expensetracker-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.3

Kubernetes:


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Images Give feedback
Local My Hub
0 Bytes / 2.92 GB in use 5 images
Last refresh: 16 hours ago
Search
Name Tag Image ID Created Size Actions
kindest/node v1.34.3 08497ee19eac 4 months ago 1.35 GB
docker/desktop-cloud-provider-kind v0.5.0 4ad59ce20658 4 months ago 595.05 MB
envoyproxy/envoy v1.36.4 08acc53db6f4 4 months ago 247.75 MB
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
shashmaran/java-app latest ff03525fd53e 4 minutes ago 675.06 MB
Showing 5 items
Engine running
Kubernetes running RAM 2.69 GB CPU 1.44% Disk: 5.75 GB used (limit 1006.85 GB)
Terminal
Update available


Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\shash> kubectl get pods
NAME READY STATUS RESTARTS AGE
app-service-ddb7b479f-m4zpg 0/1 CrashLoopBackOff 24 (5m5s ago) 11h
expense-tracker-54f9d6bb58-cl9zt 0/1 ImagePullBackOff 0 10m
quiz-app-5d49566f5f-zk2lr 0/1 CrashLoopBackOff 77 (4m56s ago) 15h
PS C:\Users\shash> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
app-service NodePort 10.96.127.76 <none> 8080:30081/TCP 11h
expensetracker-service NodePort 10.96.90.65 <none> 8080:30082/TCP 10m
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 15h
quiz-service NodePort 10.96.197.219 <none> 8082:30080/TCP 15h
PS C:\Users\shash>